

How do V1 neurons adapt to repeated stimuli?

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Abstract

In this paper, we investigate how neurons in the primary visual cortex (V1 area) of mice adapt to repeated presentations of identical stimuli. Through analysis carried out on the Allen Brain Observatory dataset we find statistically significant evidence that the response becomes less pronounced with repetition, however we also outline multiple limitations of these results.

Introduction

While it is widely known that neuron adaptation is a central mechanism in learning and efficient sensory coding (Kohn), it is difficult to pinpoint where adaptation is most prominent and impactful.

With this in mind, our aim is to describe the way in which V1 neurons adapt by investigating the responses of these neurons when mice are exposed to repeated visual stimuli.

Data selection and processing

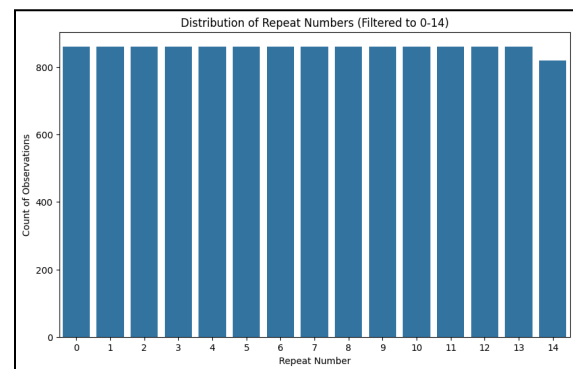
In this project, we use data from the Allen Brain Observatory ecephys project. We present results from an arbitrarily chosen session, since they are consistent with the others we analyzed.

Our first step was to isolate a type of stimulus shown during ecephys session which would be suitable for the following analysis. We chose the *drifting gratings* stimulus type. As the stimulus is described by only 3 parameters: *orientation*, *temporal frequency*, and *spatial frequency*, the data processing and analysis becomes easier.

After selecting all the stimuli of the *drifting gratings* type, we proceed by assigning a *repeat_number* variable to each presentation, representing the amount of times a grating with the exact same set of parameters as the current grating was shown in the session previously. We utilize the fact that any given set of parameters corresponds to a unique value of the *stimulus_condition_id* variable, which is provided in the dataset.

To verify that the session contains sufficient data, we graph the distribution of the *repeat_number* variable

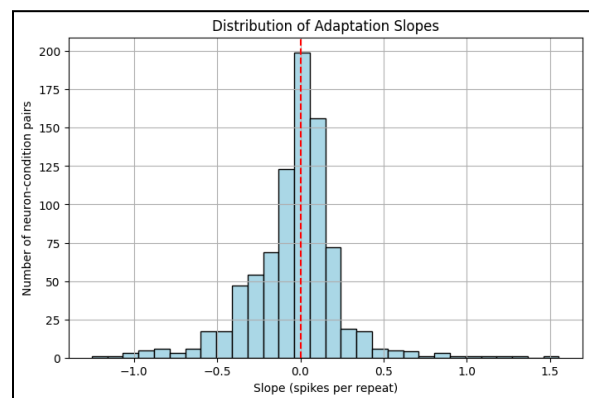
on a histogram, noticing that a great majority of the unique gratings were presented 14 times.



Once we're satisfied with the data selected, we calculate the spike counts for all V1 unit and stimulus pairs and record that information for the analysis.

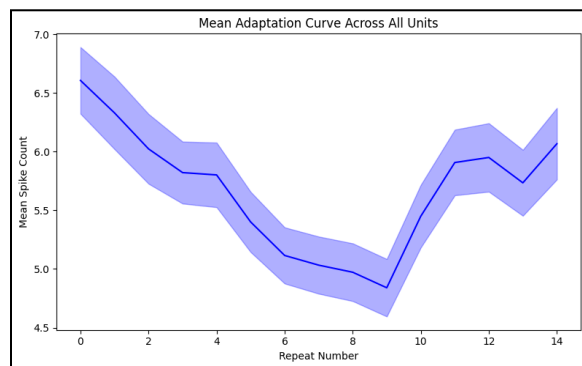
Analysis

Our first analysis step is linear regression - we fit a line to each neuron-condition pair's spike count versus its repeat number. Graphing the distribution of the slopes on a histogram we find that the slopes are close to 0, with more on the negative side.

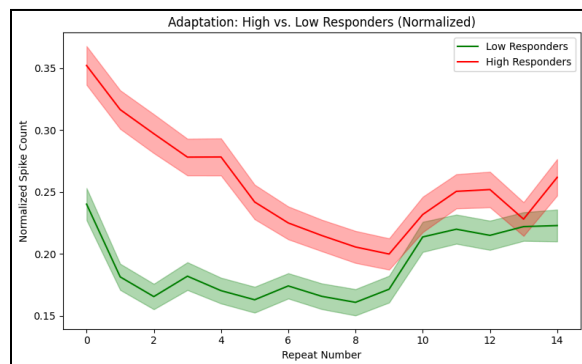


Using a Wilcoxon signed-rank test, we find a p-value of 0.00187, with a mean slope of -0.0347. Therefore, we can say with confidence that neurons show a small, negative adaptation (that is a decrease in the firing rate) when presented with repeated stimuli.

Next, we looked at the average transient response (0-500ms) across all units and stimuli. The mean spike counts dip from about 6.6 spikes on repeat 1 down to just below 5 spikes by repeat 9, and then bounce back towards 6 spikes on repetition number 14. This behavior might be explained by the network resetting when adaptation mechanisms fatigue or inhibitory circuits relaxing. (Jin and Glickfeld)



To further investigate, we wanted to check whether high and low firing neurons adapted differently. We selected units in the 1st and 4th quartiles based on the spike counts and graphed the mean normalized spike counts in these groups against the repeat numbers of stimuli. We find that both types of neurons adapt in the same way, however, the negative slope is steeper for high firing neurons.



Finally, we wanted to see whether adaptation varied based on the grating parameters. For this, we used the

Kruskal-Wallis test (a non-parametric version of a one way ANOVA) We found $p = 0.027$ for orientation and $p = 0.275$ for temporal frequency. In the case of spatial frequency, the test could not be performed due to a lack of variety in the data. The statistically significant result for orientation shows that V1 neurons adapt differently for different orientation gratings. On the other hand, the temporal frequency test shows no evidence that the speed with which the gratings move has any effect on how neurons respond to repetition.

Limitations of the approach

There are some clear limitations in our approach that we would like to mention. These limitations are:

1. lack of diversity of the analyzed stimuli
2. no analysis of adaptation in between sessions for a single specimen
3. lack of feedback to the mice during data collection

An explanation of these limitations is included in the appendix.

Conclusion

During the course of our research, we found statistically significant evidence that neurons in the primary visual cortex of mice adapt their response when presented with identical stimuli multiple times. Over the course of a data collection session used for our dataset, the neuron response becomes lower at first, and then bounces back slightly. Overall, V1 neurons fire less when they are presented with an identical stimulus repeatedly. This seems to be consistent with previously published results. (Jin and Glickfeld)

Analysis using the Kruskal-Wallis test showed that only the orientation of the presented grating has a statistically significant effect on how the neurons adapt to that stimulus.

We also present limitations in our study stemming from the nature of the dataset, as well as the analysis carried out in this paper, and present possible avenues for further research.

Appendix

Limitations of the approach expanded:

1. The results would be more comprehensive if more types of stimuli were included in the analysis. We decided to focus only on *drifting_gratings* to keep the scope of this paper focused, but this is a clear avenue for further research.
2. Analyzing multiple sessions from a single specimen would provide further data about the neuron adaptation over a larger amount of stimuli repetitions.
3. Since the mice receive no feedback during data collection, some mechanisms which cause neuron adaptation might not be triggered during data collection, which may in turn obscure a significant part of the way in which V1 neurons adapt to repeated stimuli.

References:

Jin, Miaomiao, and Lindsey L. Glickfeld.

“Magnitude, time course, and specificity of rapid adaptation across mouse visual areas.” Journal of neurophysiology, vol. 124, no. 1, 2020, pp. 245-258.

Kohn, Adam. “Visual adaptation: physiology, mechanisms, and functional benefits.” Journal of neurophysiology, vol. 97,5, 2007, pp. 3155-3164.